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# Distribution and Environmental Preference of Potential Mercury Methylators in Paddy Soils across China

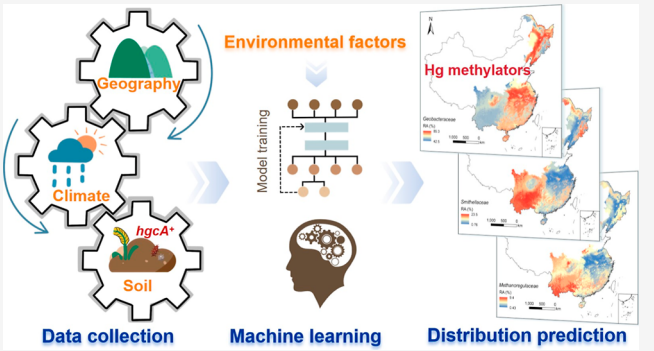
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**ABSTRACT:** The neurotoxin methylmercury (MeHg) is produced mainly from the transformation of inorganic Hg by microorganisms carrying the *hgcAB* gene pair. Paddy soils are known to harbor diverse microbial communities exhibiting varying abilities in methylating inorganic Hg, but their distribution and environmental drivers remain unknown at a large spatial scale. Using *hgcA* gene amplicon sequencing, this study examined Hg-methylating communities from major rice-producing paddy soils across a transect of ∼3600 km and an altitude of ∼1300 m in China. Results showed that *hgcA*$^+$ OTU richness was higher in tropical and subtropical paddy soils compared to temperate zones. *Geobacteraceae*, *Smithellaceae*, and *Methanoregulaceae* were identified as the dominant *hgcA*$^+$ families associated with MeHg production, collectively accounting for up to 77% of total *hgcA*$^+$ sequences. Hierarchical partitioning analyses revealed that pH was the main driver of *hgcA* genes from *Geobacteraceae* (14.8%) and *Methanoregulaceae* (16.3%), while altitude accounted for 21.4% of *hgcA* genes from *Smithellaceae*. Based on these environmental preferences, a machine-learning algorithm was used to predict the spatial distribution of these dominant *hgcA*$^+$ families, thereby providing novel insights into important microbial determinants for improved prediction of MeHg production in paddy soils across China.




**KEYWORDS:** *Hg-methylators, hgcA, distribution, environmental drivers, paddy soils*

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## INTRODUCTION

Mercury (Hg) is a global contaminant that can be transformed to the neurotoxic methylmercury (MeHg) by certain microorganisms carrying the *hgcAB* gene pair.[1,2] MeHg is of great health and environmental concern because of its tendency to bioaccumulate in aquatic and terrestrial food webs with subsequent exposure to humans and wildlife.[3] Recent studies have demonstrated that, in China, human exposure to MeHg is primarily caused by ingestion of rice grains.[4] Due to frequent flooding during the rice growing season, the paddy soil is a hotspot for microbial anaerobic processes such as Hg methylation.[5] For decades, Hg methylators (*hgcA*$^+$ microorganisms) have been identified as *Deltaproteobacteria* (reclassified recently as *Desulfobacterota*) including sulfate-reducing bacteria, iron-reducing bacteria, and syntrophic bacteria,[6] fermentative *Firmicutes*, and methylotrophic *Methanomicrobia* clade.[6−9] Recently, the discovery and study of the diversity of *hgcA* genes from environmental samples revealed the presence of potential Hg methylators within a vast diversity of other microbial taxa, such as *Chloroflexi*, *Nitrospina*, *Verrucomicrobia*, and several other phyla.[8,10,11] However, the distribution and environmental conditions governing Hg-methylating communities in paddy soils remain poorly known. Given the distinct Hg methylation potentials exhibited by different microbial taxa,[12−14] a comprehensive understanding of the diversity and composition of *hgcA*$^+$ communities is critically needed for predicting MeHg production in paddy soils across China.

Numerous studies have suggested that the diversity of soil microbial communities is regulated by the complex interplay of biogeochemical processes and environmental factors, such as geographic locations, soil properties, and climate regimes.[15−18] In particular, soil properties are considered the key drivers of microbial Hg methylation under anaerobic conditions.[19] Hg methylation can be affected by many environmental factors, such as pH, soil organic carbon (SOC), and sulfur compounds,[20] as these factors could significantly influence Hg-methylating communities.[21] For example, the widely recognized inverse relationship between MeHg production

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and pH could be related to pH impacts on Hg-methylating microorganisms,[22,23] while the SOC content and composition may influence the activity and diversity of Hg-methylating communities in distinct manners.[24,25] Moreover, spatial and climate variations are known to affect the soil microbial communities, as elevated temperature could promote microbial communities associated with Hg methylation.[19,26,27] Previous studies have shown that the diversity and geographic distribution of microbial communities in paddy soils are influenced by the mean annual temperature (MAT) and site altitude.[28,29] The soil microbiome is also influenced by multiple environmental variables, although the relative importance of environmental drivers that control the distribution of different Hg methylators remains elusive on a broad spatial scale.

To address this knowledge gap, we collected 105 composite paddy soil samples, covering >87% rice-producing areas across 3600 km from southern to northern China. These locations include soil pH ranging from 4.6 to 7.9 and altitude varying from 3 m to 1230 m. We aimed to identify the main drivers of $hgcA^+$ communities and to provide microbial parameters for the improved prediction of Hg methylation at a large spatial scale. However, the effects of microbial activity and Hg bioavailability on MeHg production were not examined, as they are not the focus of this study. To do this, we first characterized the diversity and composition of the $hgcA^+$ community using *hgcA* gene amplicon sequencing and measured the main soil variables [e.g., total Hg, MeHg, pH, SOC, total N (TN), etc.]. Then, we identified the associations between $hgcA^+$ microorganisms and 15 environmental variables relating to geographic, climatic, and soil conditions. Finally, we created a continental-scale map of the dominant microbial Hg methylators using a machine-learning algorithm.

## MATERIALS AND METHODS

**Study Areas and Soil Sampling.** Soil samples were collected from 105 typical rice paddy locations across 3600 km in China (from 18.26° N to 48.04° N, Figure S1a and Table S1), including 67 previously used samples and data sets[33] and 38 new samples collected in this study. These sites spanned tropic, subtropic, warm-temperate, and mid-temperate zones.[30] These temperature zones cover a broad range of climate and environmental gradients, with MAT ranging from 1.7 °C to 26.0 °C and annual precipitation varying from 513 mm to 2023 mm. These sampling sites also covered a diverse range of altitudes, ranging from 3 m to 1230 m (Figure S1b). Based on average altitude, the terrain of China is typically divided into three levels: the first gradient terrain (average altitude >4000 m), the second gradient terrain (average altitude 1000−2000 m), and the third gradient terrain (average altitude <500 m).[31,32] As the main rice-producing areas of China are primarily distributed in the second and third gradient terrains, the samples were collected following the same distribution pattern (Figure S1b). Further details on the principles of the above temperature zones and gradient terrain divisions are provided in the Supporting Information (Text S1). We sampled the soils right after harvest (i.e., within a week) when the conditions of the paddy fields were comparable to those during rice growth. Moreover, post-harvest soil sampling is a standard method for the analysis of functional genes, as genomic DNA residues exhibit stability in the soil for several months.[33] We collected five soil cores per plot at depths of 0−15 cm at each site. Soil samples from five individual cores were mixed thoroughly to generate a final composite soil sample.[34] All soil samples were placed on dry ice immediately and transported to the laboratory within 24 h. Subsequently, the sample was divided into two subsamples, one was stored at −20 °C for microbial analysis, and the other was air-dried and then sieved through a 2 mm mesh for soil chemical analysis.

**Analyses of Hg-Methylating Communities.** For each of the 105 composite soil samples, DNA was extracted from 0.25 g of freeze-dried soil using a DNeasy PowerSoil Kit (Qiagen, Benelux BV; previous PowerSoil DNA Isolation Kit from Mo Bio Laboratories) according to the manufacturer's instructions. The primers *hgcA*_261F (5′-CGGCATCAAYGTCTGGT-GYGC-3′) and *hgcA*_912R (5′-GGTGTAGGGGGTG-CAGCCSGTRWARKT-3′) were used to amplify a portion of the *hgcA* genes.[35,36] The PCR reaction mixture including 4 μL of 5× FastPfu buffer, 2 μL of 2.5 mM dNTPs, 0.8 μL each primer (5 μM), 0.4 μL of FastPfu polymerase, 10 ng of template DNA, and $ddH_2O$ to a final volume of 20 μL. The amplification conditions consisted of an initial denaturation at 98 °C for 2 min, followed by 35 cycles of 10 s at 96 °C, 30 s at 58 °C, and 45 s at 72 °C, and terminated by a final 7 min extension at 72 °C. The PCR products were purified using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA) and quantified using Quantus Fluorometer (Promega, USA). Purified amplicons were sequenced using Illumina MiSeq (Illumina Inc., San Diego, CA, United States) with a 2 × 300 bp paired-end protocol at Majorbio in Shanghai, China. Low-quality sequences were trimmed with SICKLE,[37] and read adapters were removed with CUTADAPT.[38] The fragment length of the *hgcA* gene (651 bp) exceeds the available sequencing length (600 bp) and led to difficulties in sequence assembly, and we thus performed artificially linking the forward and reverse reads with "N" bases. The obtained reads were truncated, dereplicated, and singletons were removed using USEARCH version 8.0.[39] Clustering of reads into operational taxonomic units (OTUs) was based on 85% similarity threshold.[13] A hidden Markov model built for *hgcA* gene was used to search with HMMER[9] *hgcA* sequences for taxonomical annotation with a manually curated database of Hg-MATE-Db.[9,40] Sequence hits with an *E*-value cutoff score of $<1 \times 10^{-5}$ were removed to ensure high confidence in all hits.

**Soil Analyses and Collection of Environmental Variables.** We analyzed 15 environmental parameters and merged them into three categories, including geographic, climatic, and soil factors (Table S1) to identify the main drivers of $hgcA^+$ microbial distribution. Soil factors, including total Hg, MeHg, pH, SOC, DOC, TN, and $SO_4^{2-}$ were measured in the laboratory. For total Hg analysis, soil was digested by a mixed solution (2 mol $L^{-1}$ $HNO_3$ and 4 mol $L^{-1}$ HCl) and determined via cold vapor atomic fluorescence spectrometry (CVAFS).[41] The detection limit was 0.2 ng $L^{-1}$, and the recoveries of the standard reference material (GBW07408 GSS-8) ranged from 86.3% to 109.5%. For the MeHg analysis, soil samples were digested with nitric acid (25%, v/v) and $CuSO_4$ (2 mol $L^{-1}$), followed by the extraction with $CH_2Cl_2$ and ultrapure water (reverse extraction), prior to the measurement of MeHg by CVAFS (Brooks Rand Model III, Brooks Rand Laboratories, USA).[4] The analytical blank for MeHg was lower than the detection limit (0.002 ng $L^{-1}$), and the recoveries of the standard reference material (ERM-CC580) ranged from 81.5% to 113.6%. Soil pH (extracted using ultrapure water, water−soil ratio of 2.5:1) was


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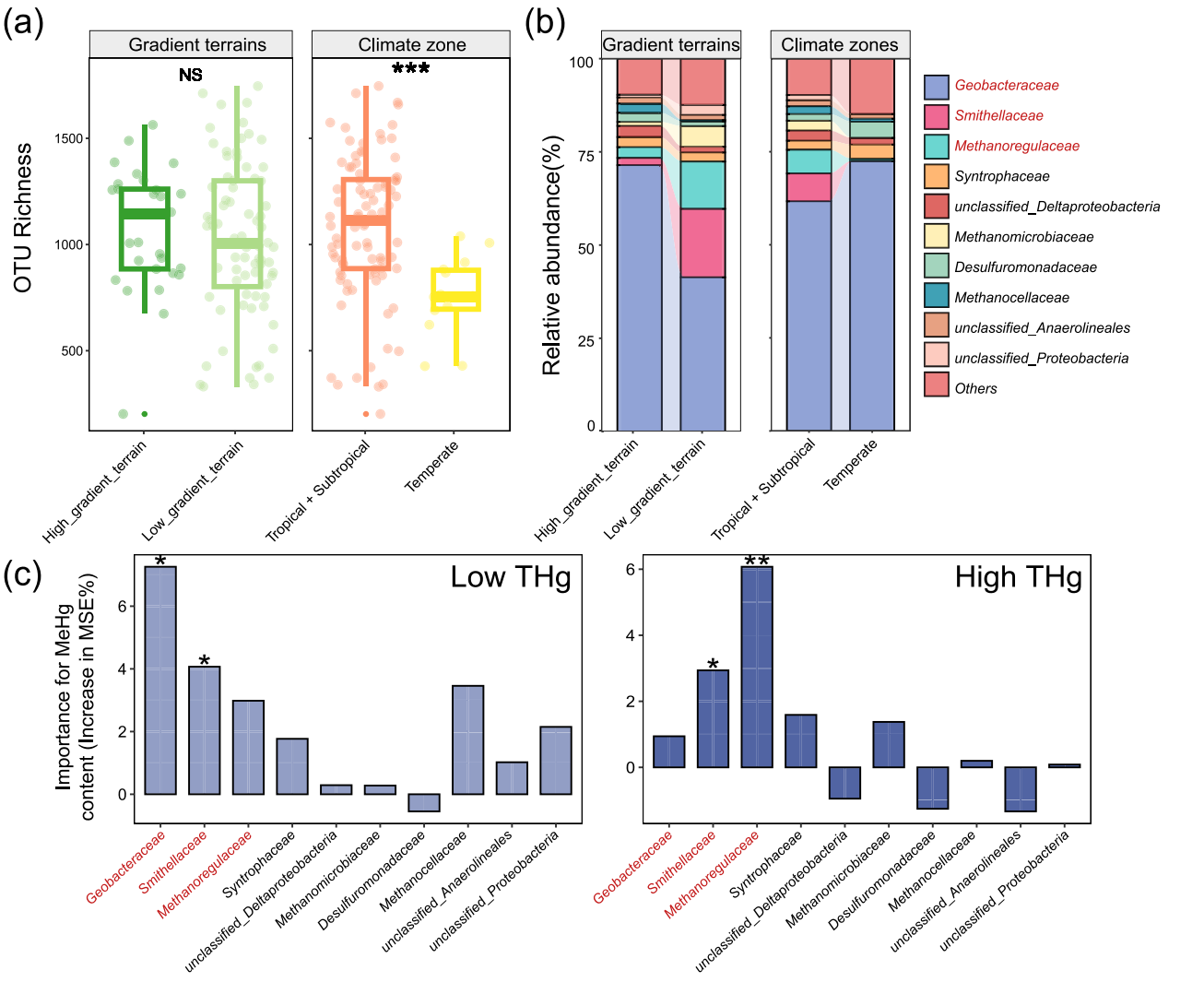



**Figure 1.** (a) OTU richness of *hgcA*$^+$ microorganisms in different gradient terrains and temperature zones across main rice-production areas of China. (b) Composition of Hg-methylating community at the family level in different gradient terrains or temperature zones. (c) Importance of three dominant Hg-methylating taxa in Hg methylation estimated by random forest model. Soil samples were collected from the second gradient terrain with relative higher altitude (ranging from 1000 to 2000 m) and the third gradient terrain with relative lower altitude (average altitude <500 m) in China (details on gradient terrain grouping are shown in Supporting Information). *$p$ < 0.05; **$p$ < 0.01; and ***$p$ < 0.001, NS: no significant.

determined by a pH meter (Mettler Toledo Instruments Co. Ltd.). SOC was analyzed via a $K_2CrO_7$ oxidation titration method. Soil TN was measured with an elemental analyzer (Vario PYRO Cube).[42] Soil CN ratio was calculated by dividing SOC by TN. DOC was extracted through adding 0.5 mol $L^{-1}$ $K_2SO_4$ at a solid-to-solution ratio of 1:5 and measured by the total organic carbon analyzer (multi N/C 2000, Analytik Jena, Germany). $SO_4^{2-}$ (extracted using ultrapure water with a water/soil ratio of 5:1) was determined through ion chromatography (Metrohm 930, Switzerland). Soil clay content was obtained from the high-resolution National Soil Information Grids of China (http://soil.geodata.cn; ∼1 km resolution). Geographic variables including longitude, latitude, and altitude were recorded by a GPS device at sampling sites. Climatic variables including MAT, temperature seasonality (TS), mean annual precipitation (MAP), and precipitation seasonality (PS) were taken from the WorldClim version 2 database (https://www.worldclim.org; ∼1 km resolution).

**Statistical Analysis.** The *hgcA*$^+$ OTU richness in each sample was calculated by tallying the number of observed OTUs, where an OTU with a count above zero is accounted for as one unit. The OTU richness between different gradient terrains or temperature zones were evaluated by Kruskal−Wallis rank-sum tests. Principal-coordinates analysis (PCoA) based on Bray−Curtis dissimilarities calculated from *hgcA*$^+$ proportion values by the "vegan" R package was used to indicate the differences in *hgcA*$^+$ community structure between different gradient terrains or temperature zones. Permutational multivariate analysis of variance (PERMANOVA) was used to test the significance of the community difference using the "vegan" R package.[43] The random forest analysis was performed to identify the predictors of *hgcA*$^+$ family associated with the MeHg concentration using the "rfPermute" package in R. We applied 1 mg $kg^{-1}$ as the threshold value to group soil Hg pollution according to the Chinese soil pollution standard (GB 15618−2018),[45] and this is helpful for our understanding of Hg-methylating microorganisms in the sites with distinct levels of Hg concentration.[44] The importance of Hg methylators for MeHg concentration was estimated by comparing the percentage increases in mean squared error


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(% Inc.MSE),[46] with higher % Inc.MSE value indicating that the *hgcA*$^+$ family is the more important predictor for MeHg production. Hierarchical partitioning analyses were then performed to assess the independent effect (%) of grouped variables (geographic, climate, and soil) and every single variable (14 environmental variables) on Hg-methylating community composition (represented by main axis of PCoA1) and distribution of dominant Hg methylators, which were conducted using "rdacca.hp" package.[47] The relationships between the Hg-methylating community similarity, geographic distance, and environmental similarities were evaluated with the partial Mantel test using the mantel function in the "ecodist" package. To distinguish the relative impact of environmental variables (geography, climate, and soil) on Hg-methylating community, multiple regressions on matrices (MRM) was performed using the "ecodist" package.[48] Furthermore, correlations between Hg methylators and environmental variables were calculated by Spearman's rank in the "psych" package. These statistical analyses were conducted using R 4.2.1 (http://cran.r-project.org/).

Structural equation modeling (SEM) was used to identify the direct and indirect associations of environmental variables with *hgcA*$^+$ community structure (represented by the main axis of PCoA1) and *hgcA* gene proportions of the three dominant families. A conceptual model of hypothetical relationships was constructed based on prior knowledge, assuming that the response of geography-induced variations in climate would directly influence the *hgcA*$^+$ community or indirectly through altering soil properties (Figure S2 and Table S2). As some of the introduced variables were not normally distributed, the probability that a path coefficient differs from zero was tested using a bootstrap test rather than the classical maximum-likelihood estimation. The fit of the model was examined by chi-square test ($0 \le \chi^2/df \le 2$ and $0.05 < p \le 1$) and RMSEA ($0 \le$ RMSEA $\le 0.05$), which further confirmed the fit of model by Bollen−Stine bootstrap test (0.10 < bootstrap $p \le$ 1.00). SEM analyses were conducted using AMOS 21.0 (SPSS Inc., Chicago, IL, USA).

**Predicting Dominant Hg Methylators in Rice-Producing Areas across China.** We predicted the relative abundances of the three dominant *hgcA*$^+$ families using a random forest algorithm based on their environmental preferences. Model fitting and prediction were done using "rfPermute" package in R. Random forest algorithms generate hundreds of ensembles of decision trees to create a model that can then be used for prediction. For each decision tree, a random set of points generated from the training data is used to calculate a prediction that then contributes to the overall outcome.[49] To ensure compatibility across data sets and variables, we calculated all predicted maps at a spatial resolution of ∼1 km. We considered predictors as follows: climate (MAT, MAP, and PS), altitude, and soil variables (pH, SOC, TN, and clay). When generating predicting maps across China, information on soil properties was obtained from the National Soil Information Grids of China (http://soil.geodata.cn; ∼1 km resolution). Our models returned an $R^2$ of 0.73 for *Geobacteraceae*, 0.71 for *Smithellaceae*, and 0.58 for *Methanoregulaceae*. Further validation of model fit was done, and the resulting correlations between predicted and observed relative abundances of the three dominant families were highly significant ($p < 0.001$).

## RESULTS

**Hg-Methylating Community Composition in Paddy Soils of China.** A higher richness (up to 1000 OTUs), in terms of the number of *hgcA*$^+$ OTUs, was observed in paddy soils from tropical and subtropical compared to temperate zones, while there were no significant differences in the OTU richness between the gradient terrains (Figure 1a). Moreover, multiple regression analysis revealed that decreased *hgcA*$^+$ community structure similarity correlated with increased geographical distance in the paddy soils across China ($p < 0.0001$) (Figure S3). The Hg-methylating community composition in different gradient terrains or temperature zones was significantly separated (PERMANOVA, $p < 0.001$) (Figure S4). Overall, *Desulfobacterota* showed the highest proportions of *hgcA*$^+$ OTUs (Figure S5), particularly *Geobacteraceae*, accounting for up to 63%, followed by *Smithellaceae* (6.5%) and *Methanoregulaceae* (5.6%) (Figure 1b). Soils from the tropical and subtropical zones showed higher proportions of *Methanoregulaceae* than that of the temperate zone (Figure 1b), whereas the second gradient terrain with a relatively higher altitude exhibited greater proportions of *Smithellaceae* and *Methanoregulaceae* than the third gradient terrain (Figure 1b). Random forest analyses indicated that the proportions of *Geobacteraceae* and *Smithellaceae* were important predictors for MeHg production in paddy soils with low Hg background values, whereas *Methanoregulaceae* and *Smithellaceae* were important indicators of MeHg production in areas with high Hg background values (Figure 1c).

**Environmental Factors Associated with the Distribution of Dominant *hgcA*$^+$ Family.** Hierarchical partitioning analyses revealed that soil factors explained higher proportions of the *hgcA*$^+$ community variations than geographic and climatic factors (Figure 2a). Particularly, pH explained 25.2% of the variation on the *hgcA*$^+$ community structure, while total Hg content was 4.5%. Consistently, multiple regression on matrix analyses also demonstrated that pH had a larger contribution to explain variations of *hgcA*$^+$ community structure similarity ($p < 0.0001$) (Figure S6 and Table S3).

Hierarchical partitioning analyses also revealed that soil properties explained higher variations in the *hgcA* gene proportions of three dominant *hgcA*$^+$ families than geographic or climatic variables (Figure 2b−d). The *hgcA* gene proportion of *Geobacteraceae* was predominantly influenced by the altitude (21.2%), pH (14.8%), and DOC (13.5%) (Figure 2b). Similarly, the proportion of the *hgcA* gene from *Smithellaceae* was primarily governed by the effects of altitude (21.4%), CN ratios (15.8%), and SOC (13.7%) (Figure 2c). As for *Methanoregulaceae*, the distribution pattern of the *hgcA* gene was primarily influenced by the effects of longitude (22.1%), pH (16.3%), and CN ratios (15.6%) (Figure 2d). We also found that the *hgcA* gene proportions from these three families had significant environmental preferences in paddy fields at a large spatial scale (Figure 2e). The proportion of the *hgcA* gene from *Geobacteraceae* was positively correlated with TS, while negatively correlated with altitude, total Hg, pH, SOC, and DOC. The proportions of both *Smithellaceae* and *Methanoregulaceae hgcA* genes were positively correlated with altitude, total Hg, SOC, TN, DOC, and MAP, while negatively correlated with TS. Random forest analyses also suggested that altitude was an important predictor for the three *hgcA*$^+$ families, while pH, SOC, DOC, TN, and C/N ratio were


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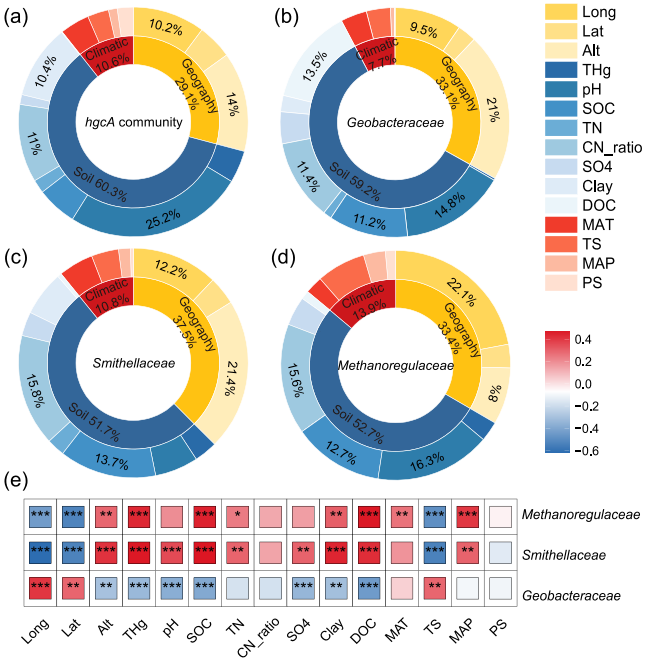



**Figure 2.** Environmental preferences of *hgcA*$^+$ community in paddy soils across China. Hierarchical partitioning analyses illustrate the importance of environmental factors affecting *hgcA*$^+$ community structure and the *hgcA* gene proportions of *Geobacteraceae* (a), *Smithellaceae* (b), and *Methanoregulaceae* (c). (e) Heatmap showing Spearman correlations between proportions of the three dominant *hgcA*$^+$ families and environmental variables. Long, longitude; Lat, latitude; Alt, altitude; MAT, mean annual temperature; TS, temperature seasonality; MAP, mean annual precipitation; PS, precipitation seasonality; *$p < 0.05$; **$p < 0.01$; and ***$p < 0.001$.

important predictors of *hgcA* gene proportions for *Geobacteraceae* and *Smithellaceae* (Figure S7).

Additional SEM analyses illustrated that climate variations along geographic locations were indirect driving forces, which influenced the *hgcA*$^+$ community structure (Figure S8) and proportions of dominant families (Figure 3) through altering soil parameters (Figure S9). Soil properties showed direct associations with the *hgcA*$^+$ community structure and proportions of the three dominant *hgcA*$^+$ families. Specifically, pH had a direct association with the proportions of *Geobacteraceae* and *Methanoregulaceae*, whereas the C/N ratio had a direct association with *Smithellaceae*. Moreover, altitude had a significant standardized total effect on both the *hgcA*$^+$ community structure and the *hgcA* gene proportions of two dominant families (i.e., *Geobacteraceae* and *Smithellaceae*).

**Spatial Distribution of Dominant Hg Methylators in Paddy Soils.** Environmental parameters were used to predict the dominant *hgcA*$^+$ family distribution across the main rice-producing areas of China using a machine-learning algorithm (i.e., random forest modeling) (Figure 4). The distribution of the three dominant *hgcA*$^+$ families in the main rice-producing areas of China varied greatly due to the distinct environmental conditions of their respective locations. For example, *Methanoregulaceae* was more prevalent in the subtropical zones of the southern regions than in the temperate zones of the northern regions. In addition, the three dominant *hgcA*$^+$ families show distinct prevalence across eastern to western China. The *hgcA*$^+$ *Geobacteraceae* was higher in lower gradients of eastern China (average altitude < 500 m), whereas *hgcA*$^+$ *Smithellaceae* and *Methanoregulaceae* showed a higher abundance in the second gradient terrains of western China (average altitude 1000 m−2000 m). Overall, *hgcA*$^+$ *Smithellaceae* and *Methanoregulaceae* posed similar spatial distribution patterns, opposite to the geographic distribution of *hgcA*$^+$ *Geobacteraceae*.

## DISCUSSION

Rice consumption is one of the main routes of MeHg exposure to humans in China,[50] and understanding microbial Hg methylation in paddy soils is thus particularly important for predicting MeHg production and assessing the potential health risks. However, insufficient knowledge regarding the prevalence of potential Hg-methylating microorganisms as bioindicators hampers the prediction of the MeHg production in paddy soils. In this study, we identify potential Hg methylators and the key environmental factors associated with their distributions over a wide range of geographic scales in China. Using this information, we were able to create predictive maps of the dominant Hg methylators for the assessment of MeHg production in paddy soils across China.

Our analyses revealed high OTU richness of *hgcA*$^+$ microorganisms across different sites, attributable to the functional versatility of Hg methylators, as they contribute to multiple metabolisms such as carbon fixation, sulfur reduction, iron reduction, and other microbial processes.[51−53] In our study, dominant *hgcA*$^+$ *Desulfobacterota* phylum, that were *Geobacteraceae* and *Desulfobacterales*, were found to be similar to those reported in other environments such as wetland sediments and seawater.[9,54] However, the predominant taxa within *Desulfobacterota* varied across different habitats. *Geobacteraceae* was the most abundant in paddy soils, consistent with that in wetland sediments,[55,56] while *Desulfobacterales* and other sulfur-reducing bacteria were more abundant in seawater.[13,57] Notably, our observations are different from previous studies showing that *Desulfobacterales* and *Desulfuromonadales* are the dominant *hgcA*$^+$ taxa in paddy soils.[36,58] This discrepancy is likely caused by the low sequencing depth by the clone library profiling and outdated databases used in those previous studies, leading to biased taxa annotation, compared to our use of the up-to-date *hgcA* database HgMATE.[40] Our study also identified *Geobacteraceae*, *Smithellaceae* (belong to *Syntrophales*), and *Methanoregulaceae* as the most abundant *hgcA*$^+$ families with high methylation potentials, as previously described.[40,59,60] A recent study also found that inhibition of *Geobacteraceae* could significantly decrease MeHg formation in paddy soils,[33] whereas methanogens could potentially regulate MeHg formation around mining areas.[61] These *hgcA*$^+$ taxa exhibit distinct Hg methylation abilities, and variations in their proportions across large spatial scales due to environmental influences can thus result in differences in MeHg formation. Therefore, identifying their spatial distribution and environmental drivers of Hg methylators is essential for improved prediction of MeHg production in paddy soils across different regions at varying temperatures and altitudes.

Compared to geographic or climate factors (Figure 2), our multianalyses indicate an overriding role of soil properties as one of the main drivers of *hgcA*$^+$ community, as previously observed.[62] Of particular importance is soil pH, which has been considered the determining factor for the Hg-methylating community. Microbial enzymatic activity is generally effective within a specific pH range.[63] In addition, pH regulates the availability of essential electron donors for microbial


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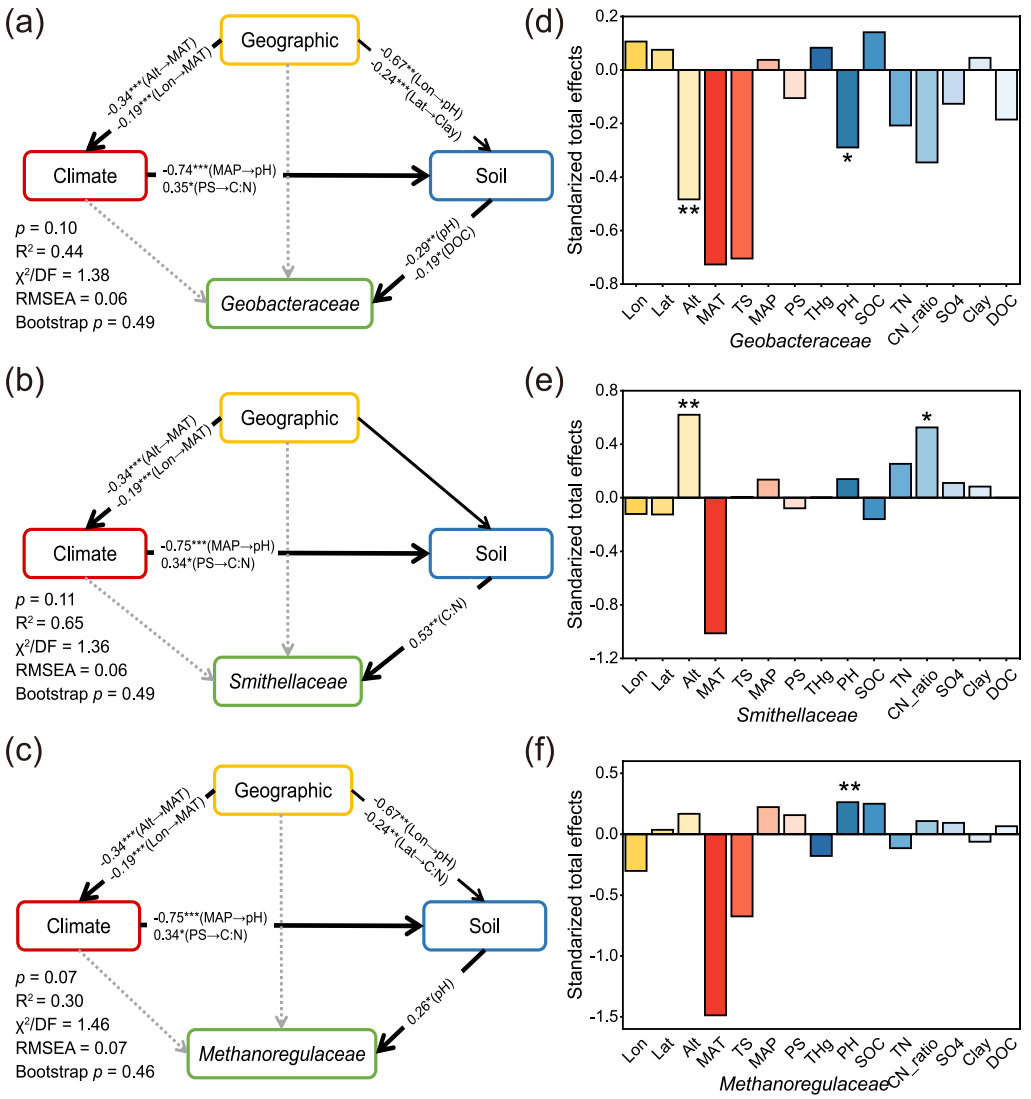



**Figure 3.** Structural equation models (SEMs) showing the pathways of environmental factors influencing the *hgcA*$^+$ community. Direct and indirect associations of geographic, climate, and soil factors with the *hgcA* gene proportions of *Geobacteraceae* (a), *Smithellaceae* (b), and *Methanoregulaceae* (c). The standardized total effects (sum of the direct and indirect effects, STE ± bootstrap confidence interval 95%) of different factors on the *hgcA* gene proportions of *Geobacteraceae* (d), *Smithellaceae* (e), and *Methanoregulaceae* (f). Different variables of each environmental factor (geographic, climate, and soil properties) were grouped in the same box for the simple illustration of SEM. The width of arrows is proportional to the potential causal effects between variables, and standardized path coefficients are shown by numbers adjacent to arrows. Black solid and gray dashed arrows indicate significant ($p < 0.05$) and insignificant relationships, respectively. The prior SEM model is shown in Figure S2. *$p < 0.05$; **$p < 0.01$; and ***$p < 0.001$.

metabolisms.[64] For example, there was a negative correlation between pH and the relative abundance of *Geobacteraceae*, and this might be due to high pH, which reduced the availability of its electron acceptor (e.g., $Fe^{3+}$). In addition, the C/N-related properties (e.g., SOC, DOC, TN, and C/N ratio) had high explanation for variations in the relative abundance of the three domain *hgcA*$^+$ families, probably due to the fact that they can utilize various soil substrates as energy sources or electron donors.[24,25,65] Overall, both pH and CN-related soil properties are main factors in affecting Hg-methylating community.[24,66] However, our multiple analyses indicate that the total Hg content had a weak association with the relative abundance of *hgcA*$^+$ microbes, and this could be explained by RNA-based *hgcA* gene expression in a future study. The diversity of *hgcA*$^+$ microorganisms could also be related to the horizontal gene transfer.[67] Other studies suggested that Hg methylators exhibited specific metabolic capacities including arsenic toxicity responses, sulfate reduction, fermentation, antibacterial function, and hydrogen oxidation[53,68,69] and may be traced back to the last common ancestor of bacteria and archaea.[70]

Significant geographical differences in the distribution of the three dominant *hgcA*$^+$ families are also observed across gradient terrains or temperature zones (Figures 1b and 4), suggesting that the changes in climate or soil properties induced by the geographical variations have substantial impacts on the distribution of dominant Hg methylators.[42,71] For example, *Smithellaceae* and *Methanoregulaceae* consistently preferred to inhabit the second gradient terrain areas of western China with relatively high altitude. This observation could be attributed to ample available SOC substrates (Figure S10),[42] thereby supporting their syntrophic metabolisms in paddy soils.[72,73] A previous study indicated that *Smithellaceae* is always isolated from methanogenic coculture environments.[74] Moreover, microorganisms belong to *Smithellaceae* family


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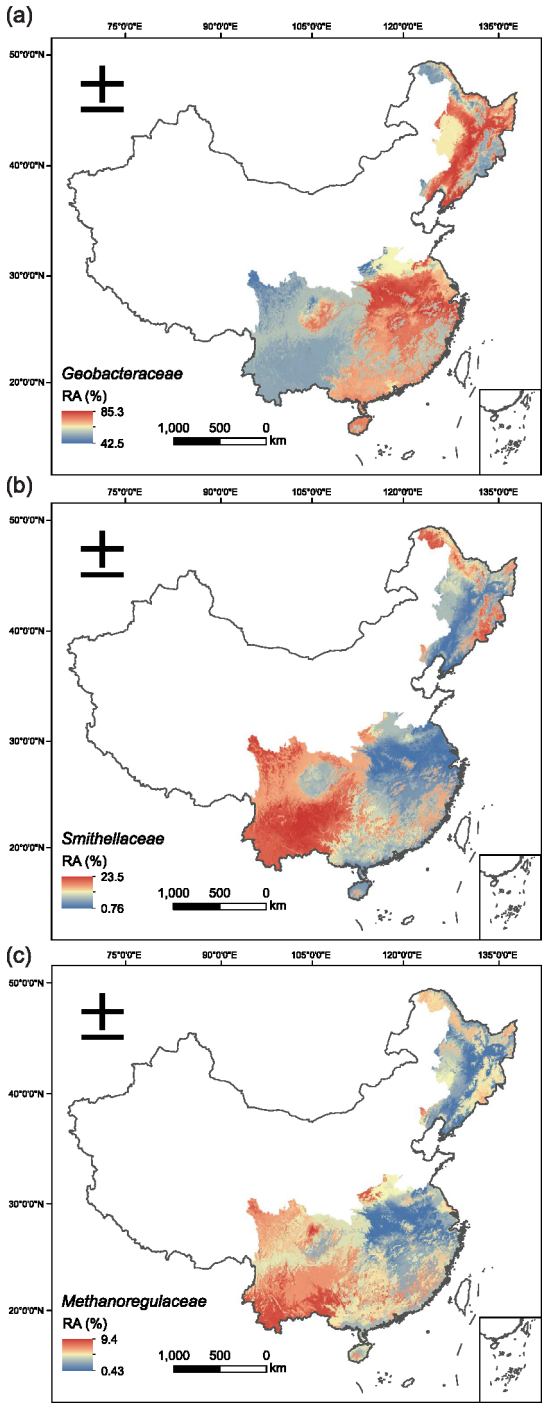



**Figure 4.** Predicted distributions of three dominant *hgcA*$^+$ families in paddy soils across main rice-producing areas of China. The explainability of model variabilities as follows: (a) proportions (RA % in the caption) of *Geobacteraceae*, $r^2$ = 0.73; $p$ < 0.001. (b) Proportions of *Smithellaceae*, $r^2$ = 0.71; $p$ < 0.001. (c) Proportions of *Methanoregulaceae*, $r^2$ = 0.58; $p$ < 0.001. The prediction was performed by a machine-learning algorithm (random forest modeling). The data collected in the 105 study sites were used for model training and validation, and then the unknown sites were predicted using spatial prediction variables including climatic (MAT, MAP, and PS), altitude, and soil (pH, SOC, TN, and clay) factors by the fitted model.

usually involve in syntrophic interactions,[74] potentially providing fermentation products to methanogens while benefiting from the removal of hydrogen and formate in the environment.[75,76] Therefore, the well-established syntrophic relationships between syntrophic bacteria and methanogens could be a key driving factor for their similar spatial distribution patterns.[73,77,78] *Methanoregulaceae* exhibited a relatively high abundance in southern China, likely due to favorable anoxic conditions resulting from abundant precipitation (Figure 2e and Table S1), which may alleviate the oxygen stress on anaerobic microbes.[79] Together, these results provide valuable insights into Hg methylation in different regions under distinct environmental conditions.

The multidimensional environmental variables help to predict the distribution of potential Hg methylators in these paddy soils in China, with the measured relative abundances of the three dominant *hgcA*$^+$ families showing a high correlation ($p$ < 0.001) with the predicted values (Figure S11); however, uncertainties still exist due to the high variabilities of microbial communities (Figure S12). Our predictions could not consider all environmental factors (e.g., variations in soil nutrient availability caused by changes in altitude) that may affect Hg methylators,[80] due to the lack of reliable geographic information for these environmental factors. Additionally, variations in redox conditions and sulfur/iron cycling, caused by differences in water management during different growth stages of rice, could also influence the *hgcA*$^+$ community composition.[22] We also note that the prediction of *hgcA*$^+$ microorganisms in this study was based on mathematical associations of these taxa with environmental parameters. New technologies and exact measurements of their contribution to MeHg production are needed in future studies. Collecting additional data, especially in diverse environmental settings, is warranted to further advance our understanding of the factors influencing Hg methylation and improve the prediction of MeHg production. Additionally, using advanced sequencing approaches such as shotgun sequencing or updated primers to detect *hgcAB* genes[9,14,67,81] may provide deeper insights into the characteristics of *hgcA*$^+$ communities and their roles in Hg cycling.

## ■ ENVIRONMENTAL IMPLICATIONS

Understanding distribution patterns and environmental drivers of Hg methylators in paddy soils is crucial for predicting MeHg production, bioaccumulation, and health risks associated with rice consumption. This study reveals the geographical distribution patterns of potential Hg methylators and their environmental preferences on a large spatial scale. *Geobacteraceae*, *Smithellaceae*, and *Methanoregulaceae* are identified as dominant potential Hg methylators in paddy soils across a broad range of environmental gradients. In particular, we show that soil properties such as pH and C: N ratio play vital roles in shaping microbial Hg-methylating communities. Altitude is also an important factor for Hg-methylator distribution, although its effect varies depending on the taxon identity. We generate predictive maps showing the distribution of Hg methylators, thereby providing useful information about the prevalence of dominant Hg methylation taxa at selected sites. Understanding environmental drivers of *hgcA*$^+$ microorganisms could also potentially help prediction of Hg methylators or MeHg formation in response to the global climate change.

This study constructs a conceptual parametrization of the dominant *hgcA*$^+$ microbial distribution in paddy soils across China and thus lays the foundation for the development of predictive models of MeHg production. Currently, the


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relationships between Hg-methylating communities and net MeHg production in paddy soils remain elusive, complicated by many compounding factors, such as DOM composition regulating the speciation and bioavailability of Hg, MeHg demethylation, non-Hg methylators, *hgcA* gene expression levels, etc.[22,25,54,82,83] Therefore, the lack of such information hinders the assessment of the Hg methylation potential.[14] Future efforts should be devoted to further elucidate environmental factors influencing microbial Hg methylation across a wide range of geographic regimes, thereby benefiting the prediction and possibly mitigation of MeHg formation in rice paddy soils.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.4c05242.

Dividing principles of temperature zone and gradient terrain; sampling locations; prior model for SEM analysis; distance decay of $hgcA^+$ community with geographical/pH distance; differences in $hgcA^+$ community structure; $hgcA^+$ phylum-level composition; importance of environmental factors by random forest analysis; SEM analysis of $hgcA^+$ community; environmental influence pathways on $hgcA^+$ microorganisms; SOC content; scatter plots of measured versus predicted values; uncertainty in $hgcA^+$ family predictions; sampling site information; rationale for environmental variable selection; and MRM analysis of $hgcA^+$ communities (PDF)

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### Notes

The authors declare no competing financial interest.

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